

Project Design Phase

Problem – Solution Fit

Date	28 June 2026
Team ID	23A51A4229
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	2 Marks

Solution Fit:

The Metro Ticket Booking System is a digital solution developed on the ServiceNow platform to simplify and automate the metro ticket booking process. It replaces the traditional manual ticketing system by allowing passengers to book metro tickets online through an easy-to-use Service Catalog interface. Users can choose their boarding station, destination station, journey date, and the number of passengers before submitting their booking request.

The system automatically determines the ticket fare based on predefined route and pricing information, ensuring accurate calculations and minimizing manual errors. Once the booking is completed, a request record is generated, a digital ticket is created, and a confirmation notification is sent to the passenger. The inclusion of QR code generation enables secure and contactless ticket verification at metro stations.

The application also provides administrative features for managing metro stations, updating fare details, and tracking booking statistics through reports and dashboards. By automating the complete ticket booking workflow, the system minimizes waiting time, improves operational efficiency, enhances the passenger experience, and offers a scalable solution for modern metro transportation services.

	DEFINE CS, FIT INTO CC	FOCUS ON J&P, TAP INTO BE, UNDERSTAND RC	EXPLORE AS, DIFFERENTIATE
DEFINE CS, FIT INTO CC	1. CUSTOMER SEGMENT CS - Urban daily commuters using public metro rail - IT and corporate office employees in tech hubs - Engineering and university students in metro cities - Weekend and occasional leisure travelers - Metro operations and ticketing admin staff - Senior citizens using subsidized metro passes	6. CUSTOMER CONSTRAINTS CC - Severe congestion at stations during rush hours - Queue wait times exceeding 20–30 minutes - No mobile or self-service digital ticketing - Staff manually calculate fares — prone to errors - Paper tickets lost or damaged easily - Counter-only access excludes remote passengers - No real-time fare transparency before travel	5. AVAILABLE SOLUTIONS AS CURRENT ALTERNATIVES - Physical counters with paper ticket issuance - Token vending machines (cash-only, limited locations) - Smart card top-up kiosks at major stations - UPI payment at selected counters - Manual telephone fare enquiry helpline KEY LIMITATIONS - All options require physical station presence - No end-to-end digital booking journey - No centralised booking or analytics system
	2. JOBS-TO-BE-DONE / PROBLEMS J&P - Book a confirmed metro ticket in under 2 minutes - Skip counter queues entirely on busy mornings - See exact fare before confirming a journey - Receive and store ticket digitally on phone - Book tickets from home, office, or on the go - Get instant booking confirmation with a reference 3. TRIGGERS - Running late and no time for counter queues - Visible long lines deterring station entry - Smartphone habit — expects app-based everything - Contactless preference post-pandemic - Peer usage of online transport booking apps	9. PROBLEM ROOT CAUSE RC 100% reliance on counter-based manual ticketing - No unified digital platform for metro bookings - Fare logic not automated — calculated by staff - Peak-hour volume far exceeds counter capacity - No workflow automation or self-service tooling - Siloed systems with no booking data or analytics - IT infrastructure not yet leveraged for passenger ops	7. BEHAVIOUR BE PATTERNS & PREFERENCES - Default to mobile for all daily transactions - Expect UPI, wallet, or card payment options - Demand live status updates and notifications - Prefer zero-human-interaction self-service flows - Share and compare digital service experiences socially - Abandon services that require extra steps or delays
EXPLORE ONLINE & OFFLINE CHANGE	4. BEFORE & AFTER (TR & EM) TR & EM BEFORE — PAIN STATE - Frustrated waiting 20+ minutes at counters - Anxious about missing a train due to queue - Uncertain about fare — has to ask staff AFTER — VALUE STATE - Carrying and managing paper tickets - Books ticket in <2 min from any location - Arrives at platform relaxed and on time - Fare shown upfront before confirming - Digital QR ticket stored in email inbox	10. YOUR SOLUTION SL - Metro ticket booking via ServiceNow Service Catalog item - Automated fare engine — calculates by source & destination - REQ/RITM ticket generated instantly on submission - Unique QR code embedded in fulfilment email - Station staff validate QR at gate — no paper needed - Live dashboard for booking volume and peak analytics - Admin portal to manage routes, fares, and capacity	8. CHANNELS OF BEHAVIOUR CH ONLINE CHANNELS - ServiceNow Service Portal (primary booking UI) - Desktop & mobile web browsers - Automated email with QR ticket attachment - SMS booking confirmation (optional) OFFLINE / PHYSICAL CHANNELS - QR scanner gates at metro entry points - Assisted booking kiosks at major stations - Metro helpdesk for booking support - Counter fallback for non-digital users